

Vol 7 Chapter 11 — Plasma and Magnetohydrodynamics

Keeper (author pass — deep math/physics revision)

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Chapter 11 — Plasma and Magnetohydrodynamics

Level 1 — one sentence

Plasma — ionized gas, the fourth state of matter and the most common form of ordinary matter in the universe — couples mechanics to EM through the Lorentz force on charged particles, with the MHD (magnetohydrodynamic) approximation treating plasma as a conducting fluid, supporting characteristic Alfvén waves and magnetic reconnection that explain phenomena from solar flares to fusion reactors.

Level 2 — graduate-physicist precision

11.1 What plasma is

A plasma is an ionized gas containing roughly equal numbers of positive ions and negative electrons (quasineutral). Plasma forms when: - Temperature is high enough to ionize: $k_B T \gtrsim$ ionization energy ($\sim$ eV for hydrogen) - Density is low enough that ionization survives recombination

Plasma is the most abundant form of ordinary matter in the universe: stars (interior + corona), interstellar medium, intergalactic medium, accretion disks, lightning, fluorescent lamps, fusion reactors.

Key plasma parameters: - Plasma frequency: $\omega_p = \sqrt{ne^2/(\epsilon_0 m_e)}$ (response timescale) - Debye length: $\lambda_D = \sqrt{\epsilon_0 k_B T / (ne^2)}$ (screening distance) - Plasma parameter: $\Lambda = n\lambda_D^3$ (number of particles in Debye sphere)

For typical hydrogen plasma at $n = 10^{20} \text{ m}^{-3}$, $T = 1 \text{ keV}$: $\omega_p \approx 5.6 \times 10^{11} \text{ rad/s}$, $\lambda_D \approx 7 \times 10^{-5} \text{ m}$, $\Lambda \approx 10^7$ (good plasma approximation).

11.2 Single-particle motion

A charged particle in static fields: - $\vec{E}$ only: linear acceleration - $\vec{B}$ only: circular gyration in plane perpendicular to $\vec{B}$ at gyrofrequency $\omega_c = qB/m$, radius $r_L = v_\perp/\omega_c$ (Larmor radius) - Both: $\vec{E} \times \vec{B}$ drift in addition to gyration

Adiabatic invariants: - Magnetic moment $\mu = mv_\perp^2/(2B)$ — invariant if B changes slowly compared to gyration - Particle bouncing between magnetic mirrors (basis of magnetic confinement)

11.3 MHD equations

For plasma with characteristic scales large compared to λ_D and timescales long compared to $1/\omega_p$, the MHD approximation treats plasma as a single conducting fluid:

- Continuity: $\partial_t \rho + \nabla \cdot (\rho \vec{v}) = 0$
- Momentum: $\rho(\partial_t + \vec{v} \cdot \nabla) \vec{v} = -\nabla P + \vec{J} \times \vec{B}$
- Ohm's law: $\vec{E} + \vec{v} \times \vec{B} = \eta \vec{J}$ (η resistivity; ideal MHD has $\eta = 0$)
- Faraday + Ampère: Maxwell's equations (with displacement current usually negligible in MHD)
- Energy: equation of state, e.g., adiabatic $P \rho^\gamma = \text{const}$

Ideal MHD ($\eta = 0$): magnetic field is “frozen-in” to the plasma — fluid elements carry their magnetic flux as they move.

11.4 Alfvén waves

In ideal MHD, transverse magnetic-field perturbations propagate as Alfvén waves at speed:

$$v_A = B/\sqrt{\mu_0 \rho}$$

The Alfvén speed depends on magnetic field strength and plasma density. Solar corona: $v_A \sim 10^6$ m/s. Earth’s magnetotail: $v_A \sim 10^5$ m/s.

Three MHD wave modes: - Alfvén waves (transverse, incompressible) - Slow magnetosonic waves (compressible, $\vec{B}$ and density coupled) - Fast magnetosonic waves (compressible, faster than Alfvén)

Substrate reading: Alfvén waves are substrate K-type collective oscillations of the plasma’s magnetic-K-type cluster.

11.5 Magnetic reconnection

In ideal MHD, field-line topology is conserved. With finite resistivity, field lines can break and reconnect, releasing magnetic energy as kinetic + thermal:

$$\partial_t \vec{B} = \nabla \times (\vec{v} \times \vec{B}) + (\eta/\mu_0) \nabla^2 \vec{B}$$

(induction equation with resistive term).

Sweet-Parker reconnection rate: $v_{in}/v_A \sim 1/\sqrt{S}$ where $S = \mu_0 L v_A / \eta$ is the Lundquist number. Often too slow for observations.

Petschek reconnection (faster) and recent collisionless reconnection theories improve agreement.

Applications: - Solar flares (sudden release of magnetic energy in corona) - Earth’s magnetosphere (reconnection at magnetopause and magnetotail) - Tokamak disruptions (rapid reconnection events) - Fast reconnection critical to all these

11.6 Fusion plasma confinement

Two main approaches to controlled thermonuclear fusion: - Magnetic confinement (tokamak, stellarator, MTF): confine hot plasma in magnetic field at low density for long time. ITER, EAST, JET, KSTAR. - Inertial confinement (ICF, NIF): compress small DT pellet rapidly to fusion conditions. NIF crossed energy break-even Dec 2022.

Lawson criterion: $nT\tau_E > 3 \times 10^{21}$ keV·s/m³ for DT fusion. Both approaches struggle with edge instabilities, energy losses, materials degradation.

11.7 Astrophysical plasmas

Most matter in the universe is plasma: - Stellar interiors: $T \sim 10^7$ K, fully ionized H, He plasma - Solar corona: $T \sim 10^6$ K, low density - Interstellar medium: $T \sim 10^2 - 10^7$ K, low density ($\sim 1 \text{ cm}^{-3}$) - Accretion disks: very hot, magnetized; central engines of AGN, X-ray binaries - Pulsar magnetospheres: extreme magnetic fields ($\sim 10^8$ T), relativistic plasmas

Substrate reading: cosmic-scale plasmas are substrate K-type configurations at cosmological Scale 3; their MHD behavior is the substrate’s coarse-grained EM-coupled fluid dynamics.

11.8 Worked example: solar plasma

Solar corona at typical loop: $n \sim 10^{15} \text{ m}^{-3}$, $T \sim 10^6 \text{ K}$, $B \sim 10^{-3} \text{ T}$.

- Plasma frequency: $\omega_p = \sqrt{(10^{15})(1.6 \times 10^{-19})^2 / (8.85 \times 10^{-12} \cdot 9.1 \times 10^{-31})} \approx 1.8 \times 10^9 \text{ rad/s}$
- Debye length: $\lambda_D = \sqrt{(8.85 \times 10^{-12})(1.38 \times 10^{-23})(10^6) / [(10^{15})(1.6 \times 10^{-19})^2]} \approx 7 \times 10^{-3} \text{ m} = 7 \text{ mm}$
- Alfvén speed: $v_A = B / \sqrt{\mu_0 \rho} \approx 10^{-3} / \sqrt{(4\pi \times 10^{-7})(10^{15} \cdot 1.7 \times 10^{-27})} \approx 7 \times 10^5 \text{ m/s}$

Solar flare reconnection releases $\sim 10^{25} \text{ J}$ in seconds — about 10^{-6} of solar surface output per second from a localized region.

11.9 K-audit anchors

- Chapter 10 (EM in matter, dielectric response)
- Volume 6 Chapter 11 (non-equilibrium thermo; MHD as non-equilibrium fluid)
- Volume 4 (GR/Cosmology; astrophysical plasmas)

Level 3 — 5th-grader accessibility

Plasma is the fourth state of matter (solid, liquid, gas, plasma) — an ionized gas with free electrons and ions. Most of the universe is plasma: stars, interstellar gas, lightning, fluorescent tubes, fusion reactor cores. Plasma physics couples mechanics to EM via the Lorentz force on charges. MHD (magnetohydrodynamics) treats plasma as a conducting fluid; in ideal MHD, magnetic field lines are “frozen into” the fluid. Alfvén waves are magnetic-field oscillations at speed $v_A = B / \sqrt{\mu_0 \rho}$; magnetic reconnection releases magnetic energy when field lines break and rejoin — drives solar flares, geomagnetic storms, tokamak disruptions. Fusion energy requires confining hot plasma (magnetic or inertial) at Lawson criterion. In BST: plasmas are substrate K-type configurations at cosmological scale; MHD behavior is the substrate’s EM-coupled fluid dynamics at coarse-grain level.

What comes next

Chapter 12 closes Volume 7 with synthesis and bridges to other volumes.

Where to look this up

- Plasma: Krall and Trivelpiece, *Principles of Plasma Physics*; Goldston and Rutherford, *Introduction to Plasma Physics*
- MHD: Davidson, *Introduction to Magnetohydrodynamics*; Priest, *Solar MHD*
- Astrophysical plasmas: Kulsrud, *Plasma Physics for Astrophysics*
- Fusion: Wesson, *Tokamaks*; Atzeni and Meyer-ter-Vehn, *The Physics of Inertial Fusion*
- BST anchors: Chapter 10; Volume 6 Ch 11; Volume 4